

Introduction to HTML

- HTML stands for HyperText Markup Language.
- It is the standard markup language used to create and structure content on the World Wide Web.
- **HTML5** is the fifth and current major version of the HTML.
- It uses "markup" to define elements like headings, paragraphs, links, images, and more.
- These elements are represented by **tags**, which are keywords enclosed in angle brackets (e.g., `<p>` for a paragraph, `<h1>` for a main heading).
- Browsers read these tags and render the content accordingly, displaying it in a human-readable format.
- It's Not a Programming Language. It doesn't perform logic or calculations like programming languages (e.g., JavaScript, Python)

Introduction to HTML

- **Structures Content:** It defines the different parts of a webpage, organizing information into a logical hierarchy.
- **Hypertext:** The "HyperText" in HTML refers to its ability to link documents together. This is done using anchor tags (`<a>`), which are fundamental to web navigation.
- **Embeds Media:** You can embed various types of media like images (`<img>`), videos (`<video>`), and audio (`<audio>`) directly into your webpages.
- **Forms:** HTML provides elements for creating interactive forms (`<form>`, `<input>`, `<button>`) that allow users to submit data.

Types of Tags

- HTML tags can be of two types.
 - Paired tags
 - Singular tags
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- A series of several parallel white diagonal lines of varying lengths, located in the bottom right corner of the slide, extending from the bottom edge towards the right edge.

Singular Tags

- A tag is said to be a singular tag if it ,does not along with a companion tag.
- A singular tag is also called stand-alone tag . This tag does not require any companion tag.
- **For example**
 - `<BR>`

Paired Tags

- A tag is said to be a paired tag if it , along with a companion tag.
- In paired tags, the first tag (<OpenTag>) is often called the opening tag and the second tag (</CloseTag>) is called the closing tag.
- The opening tag activates the effect and the closing tag turns the effect off.

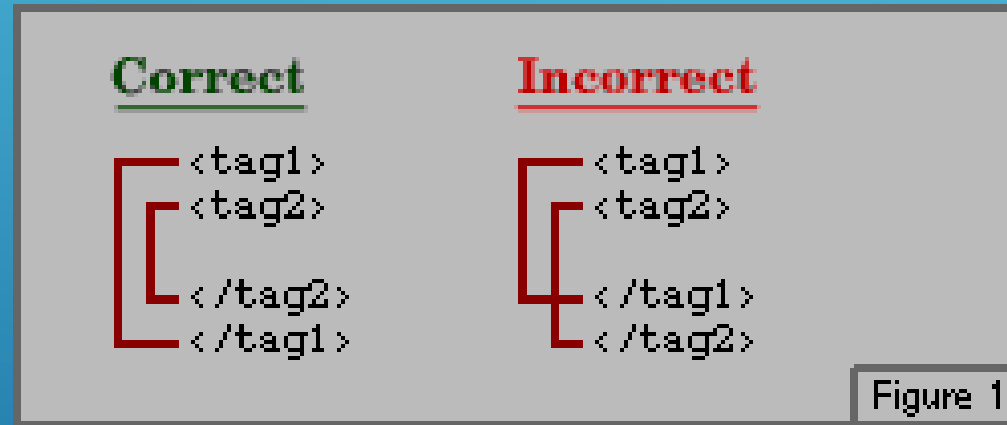
For example –

<HTML>

</HTML>

HTML Tags

- Tag-sets as containers will help in another way: it will help you remember that tags should always be balanced.



The structure of an HTML program

```
<!DOCTYPE html >
<HTML>
  <HEAD>
    <TITLE> Document Title </TITLE>
  </HEAD>

  <BODY>
    Hi!! This is my first web page.
  </BODY>
</HTML>
```

Tag Attributes

- Some HTML tags require additional information supplied to them .
- For instance, when a picture is placed on the screen, information like the height and width of the picture can be specified.
- Additional information supplied to an HTML tag is known as **attributes** of a tag.
- Attributes are written immediately following the tag in opening tag, separated by a space.
- Multiple attributes can be associated with a tag, also separated by a space.

DOCTYPE Declaration in HTML

- The HTML document type declaration, also known as DOCTYPE, is the first line of code required in every HTML document.
- The DOCTYPE declaration is an instruction to the web browser about what version of HTML the page is written in.
- This ensures that the web page is parsed the same way by different web browsers.
- By declaring **<!DOCTYPE html>** the browser to render the page according to modern, standardized rules.
- The **<!DOCTYPE html>** declaration defines that this document is an HTML5 document
- Always include it as the very first line of HTML file.

<html lang="en">

- The lang attribute is purely a declaration or a hint to browsers, search engines, and assistive technologies about what language the content is already in.
- It does not perform any translation itself.



HTML

- HTML document is mainly divided into two parts: **<head>** and **<body>**
- **<head>** : The **<head>** contains meta-information about the document.
 - ▶ Information is not directly visible on the web page, but essential for how the page works and how it's processed by browsers and search engines.
- **<body>** : The **<body>** contains everything you see on the web page: text, images, videos, buttons, forms, and more.

HTML

- **Common Uses of <head>:**
- Setting the page title (shown in browser tab)
 - ▶ **<title>My Website</title>**
- Writing CSS
 - ▶ **<style type='text/css'></style>**
- Linking CSS stylesheets
 - ▶ **<link rel="stylesheet" href="styles.css">**
- Including JavaScript files
 - ▶ **<script src="script.js"></script>**
- Metadata
 - ▶ **<meta ></meta>**

HTML <meta> Tag

- **<meta>** : This is an HTML tag used to provide metadata (data about data) about the HTML document.

These tags are placed in the <head> section of your HTML.

Setting the viewport to make your website look good on all devices:

```
<meta name="viewport" content="width=device-width, initial-scale=1.0">
```

name="viewport": This attribute specifies the name of the metadata.

It's telling the browser that this meta tag defines the "viewport" settings.

The viewport is the visible area of a web page in the browser.

HTML `<meta>` Tag

width=device-width: This tells the browser to set the width of the viewport to the actual width of the device's screen (in device-independent pixels).

So, on a phone that's 360 pixels wide, the viewport width will be 360 pixels. On a desktop screen if its 1920 pixels wide, it will be 1920 pixels.

This prevents the browser from trying to render a wider "virtual" page.

initial-scale=1.0: This tells the browser what the initial zoom level should be when the page is first loaded. 1.0 means no zoom (100%), so the content appears at its natural size relative to the device's screen.

This prevents browsers from zooming in or out by default, which can break your layout.

If I want to see the changes, I have to open the page in mobile devices.

HTML <meta> Tag

Define keywords for search engines: Keywords for search engines, though less impactful than description now).

```
<meta name="keywords" content="HTML, CSS, JavaScript">
```

Define a description of your web page: A brief summary for search engines.

```
<meta name="description" content="Page contains HTML and CSS Content">
```

Define the author of a page:

```
<meta name="author" content="John">
```

Refresh document every 30 seconds:

```
<meta http-equiv="refresh" content="30">
```

Several white lines of varying lengths and angles are positioned in the bottom right corner of the slide, creating a modern, abstract design element.

HTML Semantic Tag

Semantic tags in HTML provide meaning to the content they enclose, rather than just defining how the content should look.

This is crucial for accessibility (screen readers), SEO, and code maintainability.

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Attributes of <BODY> Tag

BGCOLOR	Changes the default background color to whatever color is specified with this tag. The user can specify a color by name or its equivalent hexadecimal number.
Exam. -	<BODY BGCOLOR = “black” > <BODY BGCOLOR = “#000000” > <BODY BGCOLOR = “rgb(0,0,0)” >
BACKGROUND	Specifies the name of the GIF or JPEG file that will be used as the background of document .
Exam. -	<BODY BACKGROUND = “abc.gif” > <BODY BACKGROUND = “../das.jpeg” > <BODY BACKGROUND = “http://wwwhpesindia.com/clouds.gif” >
TEXT	Changes the body text color from its default value to the color specified with this attributes.
Exam. -	<BODY TEXT = “red” >

Headings

- The headings are defined with the <H1> to <H6> tags.
- <H1> defines the largest heading. <H6> defines the smallest heading.
- HTML automatically adds an extra blank line before and after a heading.

<H1>Heading 1</H1>

<H2>Heading 2</H2>

<H3>Heading 3</H3>

<H4>Heading 4</H4>

<H5>Heading 5</H5>

<H6>Heading 6</H6>

Paragraphs

- Paragraphs are defined with the <P> tag.
- HTML automatically adds an extra blank line
- before and after a paragraph.

<P>This is a paragraph </P>

<P>This is another paragraph</P>

Text Formatting Tags

Formatting elements were designed to display special types of text:

- `<b>` - Bold text
- `<strong>` - Important text
- `<i>` - Italic text
- `<em>` - Emphasized text
- `<mark>` - Marked text
- `<small>` - Smaller text
- `<del>` - Deleted text
- `<ins>` - Inserted text
- `<sub>` - Subscript text
- `<sup>` - Superscript text

Character Entities

- Some characters have a special meaning in HTML, called character entities.
- If we want the browser to actually display these characters we must insert character entities in the HTML code.
- A character entity has three parts :
 - An ampersand (&)
 - An entity nameOR
 - # and entity number
 - A semicolon (;)

Character Entities

- Note that the character entities are case sensitive.
 - The most common character entity in HTML is the non-breaking space.
 - HTML will truncate spaces in text. If you write 10 spaces ,HTML will remove 9 of them .
 - To add spaces to your text, use the ** ** or ** ** character entity.
 - To display a less than sign in an HTML we must write **<** or **<**
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Character Entities

RESULT	DESCRIPTION	ENTITY NAME	ENTITY NUMBER
	Non-breaking space	 	
<	Less than	<	<
>	Greater than	>	>
&	Ampersand	&	&
“	Quotation mark	"	"
‘	Apostrophe	'	'
¢	Cent	¢	¢
£	Pound	£	£
¥	Yen	¥	¥
§	Section	§	§
©	Copyright	©	©
®	Registered Trademark	®	®
×	Multiplication	×	×
÷	Division	÷	÷
₹	Rupee Symbol		₹

Table

- A table is a two dimensional matrix, consisting of rows and columns.
 - Tables are intended for displaying data in columns on a web page.
 - All table related tags are included between the `<TABLE>` and `</TABLE>`.
 - Each row of a table is described between the `<TR>` and `</TR>`.
 - Each column of a table is described between the `<TD>` and `</TD>`.
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Table Tag

- The attributes of <TABLE> tag are :

Attributes	Description
ALIGN	Horizontal alignment is controlled by the ALIGN attribute. It can be LEFT,CENTER or RIGHT.
VALIGN	Controls the vertical alignment of cell contents. It accepts the values TOP, MIDDLE or BOTTOM.
WIDTH	Sets the WIDTH to a specific number of pixels or to a percentage of the variable screen width . The data cell is adjusted based on the cell data value.
BORDER	Controls the border to be placed around the table. The border thickness is specified in pixels.
CELLPADDING	This attribute controls the distance between the data in a cell and the boundaries of the cell.
CELLSPACING	Controls the spacing between adjacent cells.

Table Tag

- The attributes of <TABLE> tag are :

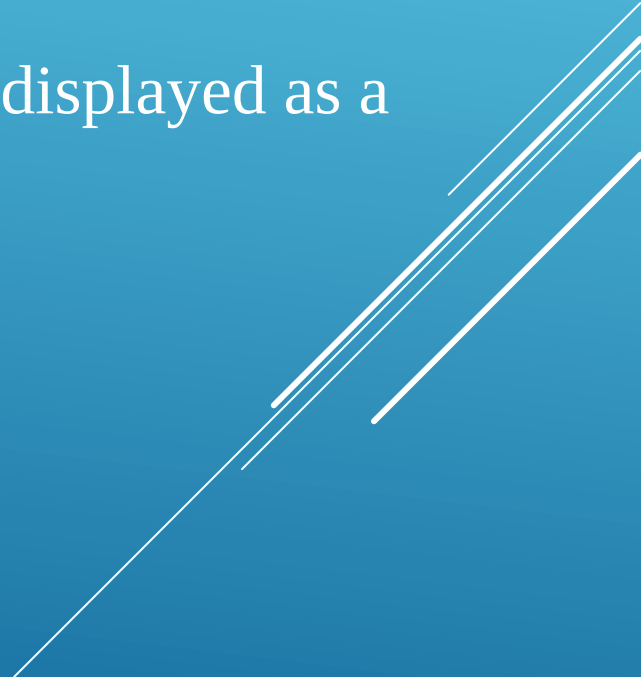
Attributes	Description
COLSPAN	The COLSPAN attribute inside a <TH> or <TD> tag instructs the browser to make the cell defined by the tag to take up more than one column. The attribute can be set by giving a numeric value . Exam. COLSPAN = 2
ROWSPAN	The ROWSPAN attribute works in the same way as the COLSPAN attribute except that it allows a cell to take up more than one row. Exam ROWSPAN = 3

Caption Tag

- Table headings are called Captions.
- Captions can be provided to a table by using `<CAPTION></CAPTION>` tags .
- This paired tag appears within the `<TABLE></TABLE>` tags.
- The table caption can be made to appear above or below the table structure with the attribute `ALIGN` .

Attribute	Description
ALIGN	It controls placing of the caption with respect to the table. ALIGN = BOTTOM ALIGN = TOP

Anchors

- Simplest possible anchor starts with `<A>` and ends with `</A>`.
 - Most anchors are in the form `<A HREF="URL">`, where URL is the location of the resource to which you want the link to point.
 - The words between the open and close of the anchor would be displayed as a hyperlink.
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Anchors

- The target attribute specifies where to open the linked document.
- The target attribute can have one of the following values:
- **_self** - Default. Opens the document in the same window/tab as it was clicked
- **_blank** - Opens the document in a new window or tab

<p> You can reach IIT at:</p>

Website

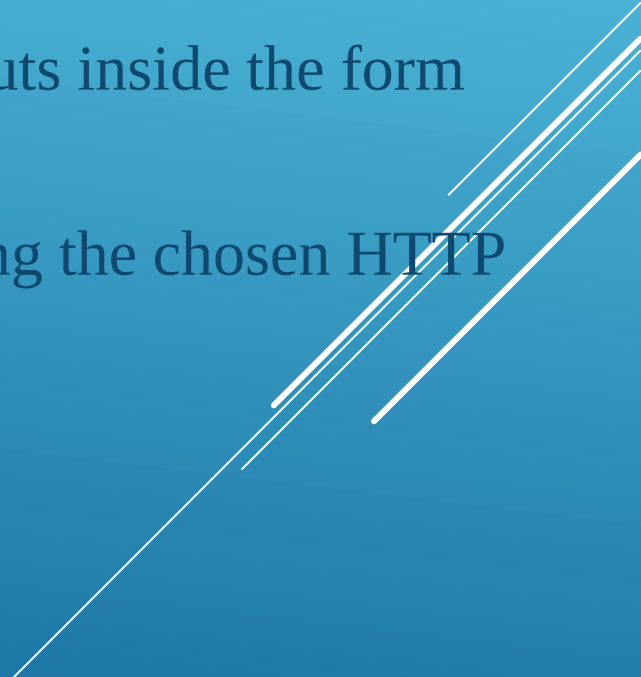
Email

Phone

submit button

- The **submit button** in HTML plays a crucial role in web forms by initiating the process of sending user-entered data to a server.
- **Event Trigger:** Clicking the submit button triggers a "submit" event on the form.
- **Data Collection:** The browser collects the name and value pairs of all successful controls within the form. Only form elements with a **name** are included in the submission.
- **Data Encoding:** The data is then encoded based on the form's enctype attribute.
- **HTTP Request:** An HTTP request (either GET or POST, as specified by the form's method attribute) is sent to the URL defined in the action attribute.

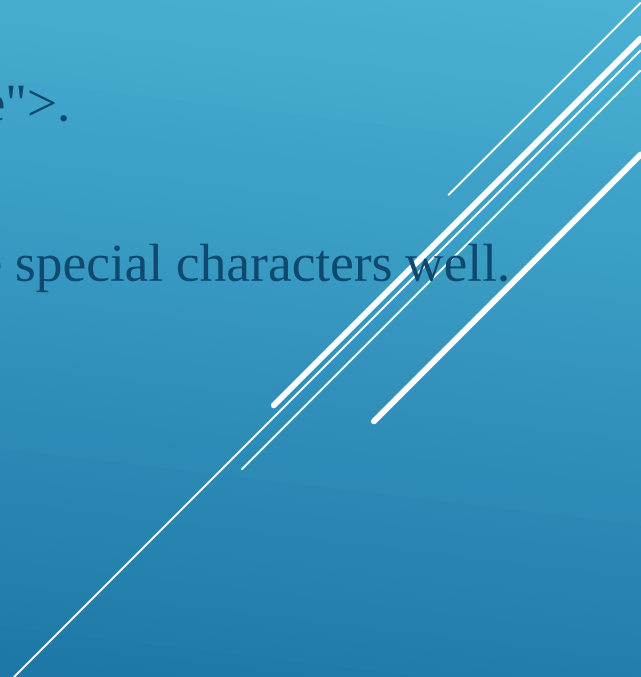
Forms

- The **<form>** element tells the browser to collect user input from its fields and send that data to a server or script for processing—such as saving to a database, performing login checks, or handling other backend tasks.
 - The form acts as a container that tells the browser what to collect and where to send it.
 - When the user clicks **Submit**, the browser gathers values from all inputs inside the form that have a name attribute.
 - Then, the browser sends those values to the specified **action** URL using the chosen HTTP method (GET or POST).
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Forms

- **action attribute:** Specifies the URL where the form data will be sent when the form is submitted.
 - This is the server-side script or endpoint that will process the data.
 - ▶ **Example:** `<form action="/process-data.php">`
- **method attribute:** Defines the HTTP(GET and POST) method to use when submitting the form data
- **GET (default):**
 - ▶ In a GET request, all form data (the key-value pairs) is appended directly to the URL as a query string (e.g. `http://example.com/search?query=myterm&category=books`).
 - ▶ Suitable for non-sensitive data or requests that retrieve data (e.g., search queries).
 - ▶ Has URL length limitations.
- **POST:**
 - ▶ Sends the form data in the body of the HTTP request (not visible in the URL).
 - ▶ Suitable for sensitive data (passwords, personal info) or large amounts of data (e.g., file uploads).
 - ▶ **Example:** `<form action="/login" method="post">`

Forms

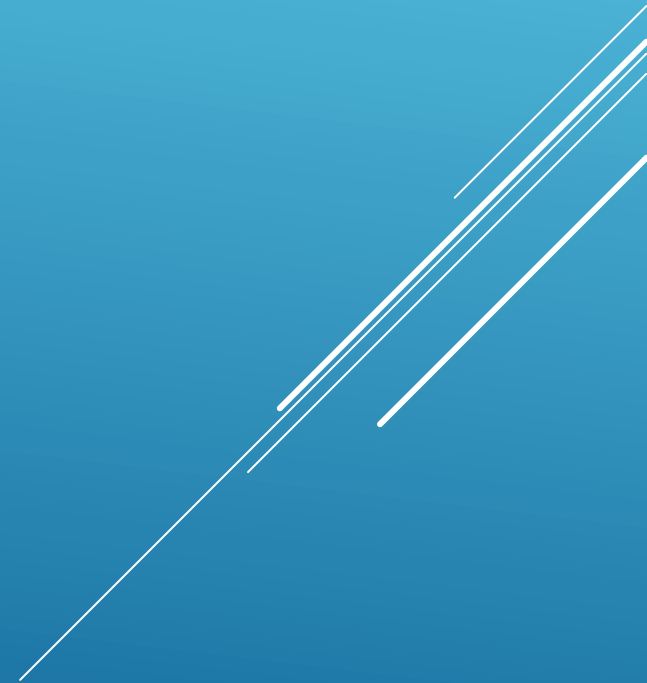
- **enctype attribute:** Specifies how the form data should be encoded when submitted (especially relevant for POST requests).
 - ▶ **application/x-www-form-urlencoded (default for GET and POST):** Encodes spaces as + and special characters as %xx.
 - ▶ **multipart/form-data:** Must be used when uploading files with `<input type="file">`.
 - ▶ **text/plain:** Sends data as plain text, primarily for debugging, as it doesn't encode special characters well.
- 

Form submission

- **The Autocomplete Attribute**

- The autocomplete attribute specifies whether a form should have autocomplete on or off.
- When autocomplete is on, the browser automatically complete values based on values that the user has entered before.

Example : `<form action="/action_page.php" autocomplete="on">`



Form submission

- **The Novalidate Attribute**
- The novalidate attribute is a boolean attribute.
- When present, it specifies that the form-data (input) should not be validated when submitted. **Example :** `<form action="/action_page.php" novalidate>`

HTML Image Maps

The **<map>** tag defines an image-map. An image-map is an image with clickable areas.

```

<map name="demomap">

<area shape="rect" coords="30,40,300,390" href="demoPage1.html">

<area shape="circle" coords="337,300,44" href="demoPage2.html">

<area shape="poly" coords="30,100, 140,50, 290,220, 180,280"
href="demoPage3.html">
</map>
```

THANK YOU

